

Heat Transfer: Hot Dog Project

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The goal of this project was to predict the time it takes for a hot dog to reach an internal temperature of 68 C without burning the surface (exceeding a temperature of 100 C) through radiative and convective heat transfer of a charcoal-burning grill. The overall approach modeled the hot dog as an infinite cylinder and the charcoal as an infinite plate. Two approaches are taken when solving for the overall time it takes to cook and the temperatures over radial position at a few time intervals. The first is an analytical approach that confirms that lumped capacitance may not be used as the Biot number is greater than 0.1. To solve analytically we account for spatial effects but use a one-coefficient approximation. Additionally, we use a 15-coefficient approximation. The second approach is a finite differencing method that takes in the temperatures at the surrounding nodes in order to calculate the temperature of a specific node at the next time step. This method is broken into the centerline temperature, the surface temperature, and all the nodes in between. MATLAB is used to iterate quickly over these solutions and make the solving process more efficient.

For calculations of a new total heat transfer coefficient, we recalculate both the convective and radiative heat transfer coefficients. These values are then plugged into Eq. 1

$$h_{total} = h_{rad} + h_{conv} \quad (1)$$

The equations to calculate each individual heat transfer coefficient are derived in further detail in Att. 3. The equations work out to the final forms seen in Eq. 2 and Eq. 3 with a resulting radiative coefficient of 5.5 W/m²-K and a convective heat transfer coefficient of 10.96 W/m²-K.

$$h_{rad} = \epsilon_1 \epsilon_2 \sigma * F_{21} (T_1 + T_2^*) ((T_1)^2 + (T_2^*)^2) \quad (2)$$

In which ϵ is the emissivity of the hot dog and the coals, σ is the Stefan-Boltzman constant, F_{21} is the view factor between the plate and the cylinder, T_1 is the coal temperature, and T_2^* is the average between the hot dog surface starting and ending temperature. The view factor scenario is determined to be a parallel plate and cylinder.

The convective heat transfer coefficient is found using the Nusselt number for flow over a cylinder and can be seen in Att. 3.

$$h_{conv} = \frac{Nu * D}{k_f} \quad (3)$$

In which Nu is the Nusselt number, D is the diameter of the hot dog, and k is the thermal conductivity of the air.

The analytical solutions were developed in Att. 1 using the assumption the hot dog can be represented by an infinite cylinder. Before solving, the Fourier number was assumed to be greater than 0.2 and was later checked after solving for the total time. First, using Eq. 4, the time

required to heat the hot dog was solved for, where θ^* is represented by Eq. 5, r^* is represented by Eq. 6, and Fo is represented by Eq. 7.

$$\theta^* = C_1 \exp(\zeta_1^2 Fo) J_0(\zeta_1 r^*) \quad (4)$$

$$\theta^* = \frac{T - T_\infty}{T_i - T_\infty} \quad (5)$$

$$r^* = \frac{r_0}{r} \quad (6)$$

$$Fo = \frac{\alpha t}{L^2} \quad (7)$$

Since the initial and final centerline temperatures of the hot dog were known with the temperature of the environment, the time elapsed could be solved for by calculating the Fourier number. For the centerline, $r^* = 0$. It was found the total time elapsed was $t = 436 \text{ sec}$. This total derived equation is shown in Eq. 8.

$$t = \frac{-1}{0.00109(\zeta_1^2)} \cdot \ln\left(\frac{1}{c_1} \cdot \frac{T_{cf} - T_\infty}{T_i - T_\infty}\right) \quad (8)$$

Next, using the total time elapsed, a time step of 0.5 seconds and a radial step of 0.0001 meters were used to determine the centerline and surface temperature of the hot dog with time. The derived equation for centerline temperature is shown below in Eq. 9.

$$T = T_\infty + (T_i - T_\infty) \cdot c_1 \exp(-\zeta_1^2 \cdot 0.00109t) J_0(\zeta_1 \cdot r^*) \quad (9)$$

For the centerline temperature $r^* = 0$ and for the surface temperature $r^* = 1$.

After updating the heat transfer coefficients, we find C1 to be 1.0936 and zeta1 to be 0.8534 with a Biot number of 0.402.

Additionally, the exact solutions for the centerline and surface time for temperature vs. time were calculated. The exact analytical solution was calculated using 15-coefficients instead of a 1-coefficient simplification. The solutions converge at the end, as can be seen in Fig. 1, but the main difference lies in the starting temperature. The derived extra credit equation for the exact temperature vs. time of the centerline and surface of the hot dog is shown in Eq. 10.

$$T = T_{\infty} + (T_i - T_{\infty}) \sum_{n=1}^{15} c_n \exp(-\zeta_n^2 \cdot 0.00109t) J_0(\zeta_n \cdot r^*) \quad (10)$$

For the centerline temperature $r^* = 0$ and for the surface temperature $r^* = 1$.

The temperature vs. time plot for the centerline and surface temperature of the hot dog for the analytical and exact solutions is shown below in Fig. 1.

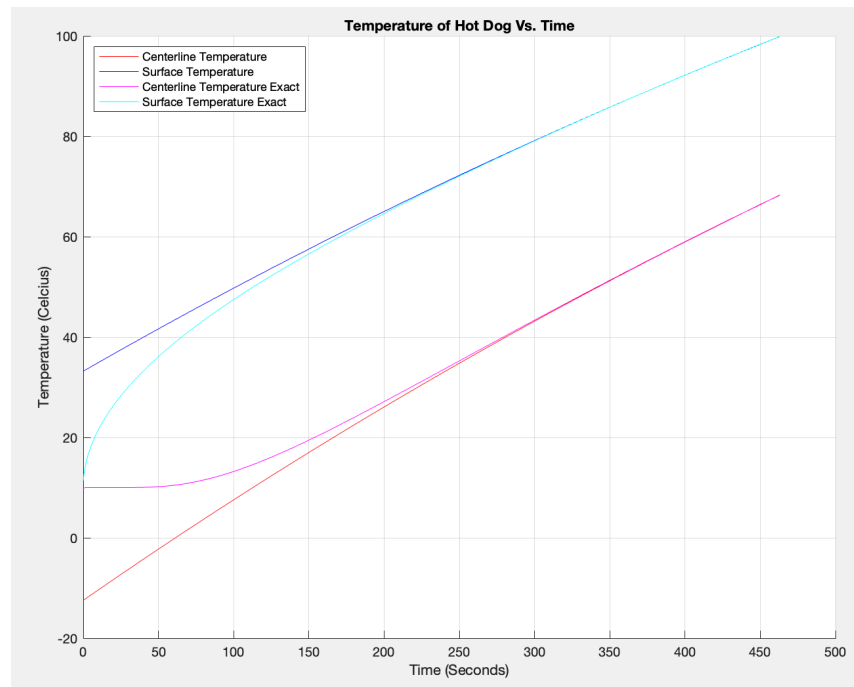


Figure 1: Temperature vs. Time for Centerline and Surface Positions.

The 1-coefficient approximation solutions are very linear and do not start at the 10 C that we know the whole hot dog starts at. The 15-coefficient solution accounts for this and we see that

both the centerline and surface temperatures start at 10 C and have a little bit more curvature as they ease into the linear trend and converge with the other solution.

The analytical solution for the temperature of the hotdog vs. position was also solved for by using Eq. 9. The difference was, in this case, t was set at intervals of $t = 30s, 120s, 436s$ and

$r^* = \frac{r}{r_0}$ where r varied radially over the entire radius of the hot dog for each time interval t .

For the exact analytical solution for the temperature of the hotdog vs. position, Eq. 10 was also used with $t = 30s, 120s, 436s$ and $r^* = \frac{r}{r_0}$ where r varied radially over the entire radius of the hot dog for each time interval t . The temperature vs. position plot for $t = 30s, 120s, 436s$, including both analytical and exact solutions is shown below in Fig. 2.

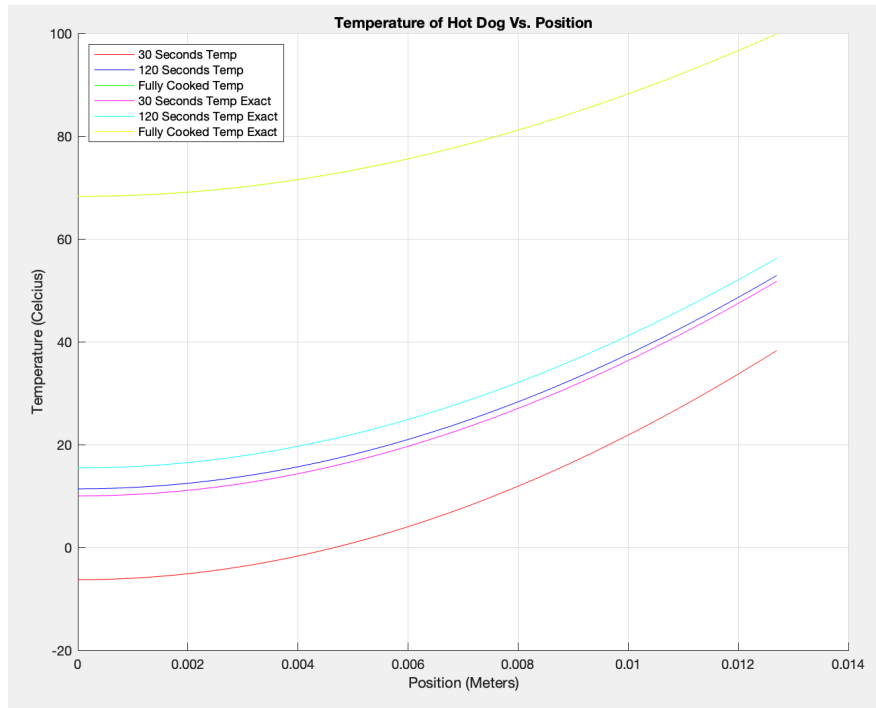


Figure 2. Analytical temperature solutions vs. radial hot dog position, both 1-term and 15-term solutions.

As seen in Fig. 2, above, the biggest difference in the 1-term and 15-term analytical solutions are the temperatures at intermediate times. This difference can be explained by the trend seen in Fig. 1. The 1-term solution has linear curves that do not start at the correct temperatures, because of this, the intermediate temperatures at specified times will be incorrect for the 1-term approximation. However, because the Fourier number was greater than 0.2 and the 1-term solution is valid, we see that the final cooked time only appears to be one line, because each solution is the same. Additionally, for the overall temperature vs. position trend, we see that it is not linear, but it is not exponential either. The temperature is the hottest on the surface and the coolest at the centerline. This makes sense as the heat is traveling inwards from the outside of the hot dog, where the source is located.

The MATLAB code for the analytical and exact solutions is in Att. 4.

The numerical finite differencing solutions were used to determine the hot dog temperature vs. time and the hot dog temperature vs. position. The development of these solutions was solved using nodal analysis. The derivation of the equations for numerical finite differencing is shown in Att. 2. For the centerline temperature, the derived equation is shown in Eq. 11.

$$T_0^{i+1} = 4F_0 T_1^i + T_0^i (1 - 4F_0) \quad (11)$$

The stability criterion for this equation is $F_0 < \frac{1}{4}$ which will be compared with the equation for interior temperatures and the equation for the surface temperature.

The equation for the interior node temperature is shown in Eq. 12. We chose the number of nodes in our hot dog to be 30 nodes, with node 1 located at the centerline and node 30 located at the surface. Equation 12 solved for nodes 2 through 29.

$$T_m^{i+1} = Fo(\frac{1}{m})\left[T_{m-1}^i(m - \frac{1}{2}) + T_{m+1}^i(m + \frac{1}{2})\right] + T_m^i(1 - 2Fo) \quad (12)$$

The stability criterion for this equation is $F_0 < \frac{1}{2}$.

The equation for the surface temperature is shown in Eq. 13.

$$T_M^{i+1} = (Fo)(T_{M-1}^i)(4 - \frac{2}{M}) + T_M^i(1 + \frac{2Fo}{M} - 4Fo) + (\frac{4\Delta t}{\Delta \rho c})h(T_\infty^i - T_M^i) \quad (13)$$

The stability criterion for this equation is $F_0 < 0.2525$.

Because the centerline temperature had the most restrictive stability criterion, $F_0 < \frac{1}{4}$, the

Fourier number used had to be less than $\frac{1}{4}$. Therefore, a timestep was chosen based on the

distance between nodes, $\frac{r}{30}$, and a Fourier number of $\frac{1}{4}$. The finite differencing derivations can

be seen in Att. 2.

The plot of the centerline and surface temperature with time for the numerical finite differencing method is shown below in Fig. 3.

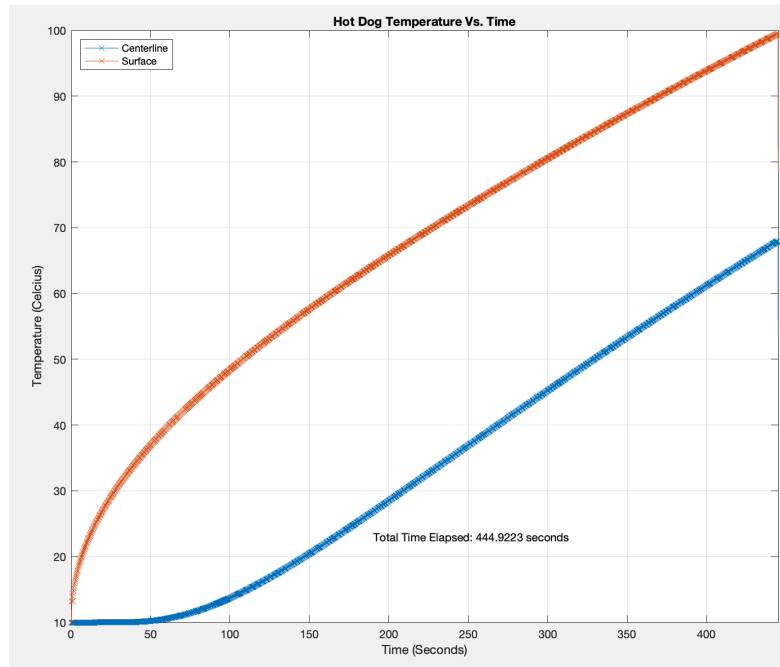


Figure 3. Finite differencing method centerline and surface hot dog temperature vs. time.

As can be seen in Fig. 3, above, both the centerline and surface temperature start at the appropriate 10 C. They then work their way into a linear temperature increase until the centerline temperature hits 68 C. When the centerline is 68 C, the surface temperature is just below 100 C. Additionally, the time step is calculated and we see that it takes around 445 seconds for the hot dog to reach 68 C internally.

The plot of the temperature vs position for $t = 30, 120$, *fully cooked* for the numerical finite differencing method is shown below in Fig. 4. Fully cooked was determined by breaking the code when the centerline temperature reached or exceeded 68°C.

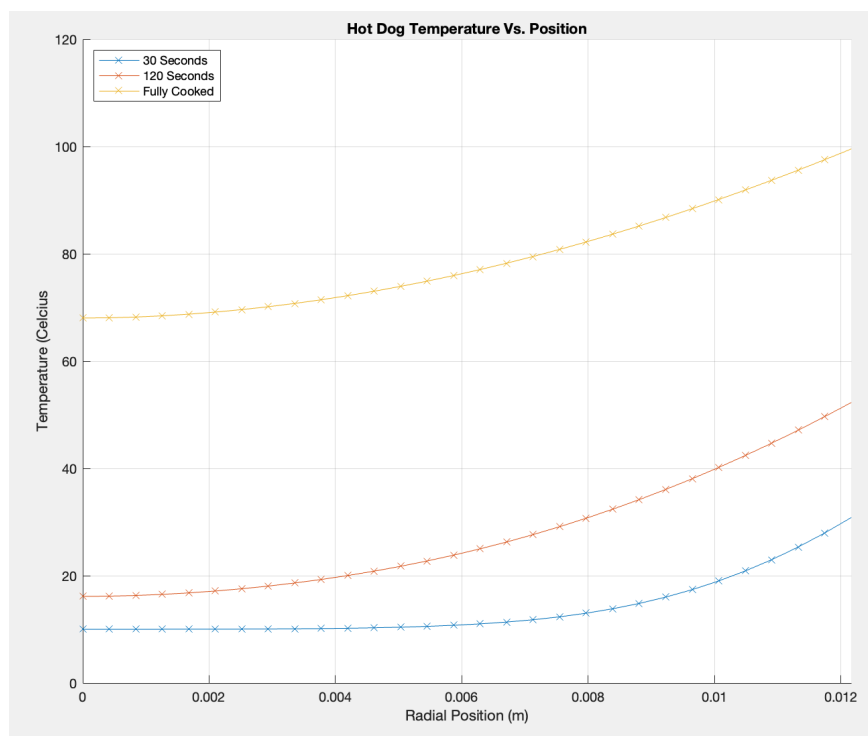


Figure 4. Finite differencing method hot dog temperature vs. radial hot dog position.

As seen in Fig. 4, the finite differencing method has fewer data points, as we only plot the specified number of nodes. For calculation purposes, the number 30 was selected as it provides a

relatively good depiction without requiring too much computational power. Because this method looks at the nodes to the left and right of the current node to calculate the temperature at the current node at the next time step, until the final cooked temperature, the nodes closest to the centerline do not see very much temperature change. This is because of the “delay” of the effect of the outer node temperatures in reaching the innermost nodes.

The MATLAB code for the numerical finite differencing solution is shown in Att. 4.

The numerical and analytical solution results were very close, we found that the analytical solution predicted a total cook time of 436 seconds, while the numerical solution predicted a total cook time of 445 seconds. When the hot dog was actually cooked we found that it took 445 seconds to reach an internal temperature of 68 C. Additionally, we see that once the heat transfer coefficient values are updated, the surface of the hot dog no longer exceeds 100 C, so it doesn't burn.

Att. 1 Analytical Solution Development

Equation Development:

→ Assume $\text{length}_{HD} \gg r_{HD} \rightarrow$ treat as an infinite cylinder

→ FROM 5.6.1/2 of textbook

$$\theta^* = \frac{T - T_m}{T_i - T_\infty} = \sum_{n=1}^{\infty} C_n \exp(-\zeta_n^2 F_0) J_0(\zeta_n r^*)$$

→ Assume $F_0 > 0.2$ to use 1-term approximation & check validity after ...

$$\theta^* = C_1 \exp(-\zeta_1^2 F_0) J_0(\zeta_1 r^*)$$

$$\text{For centerline } r^* = \frac{r}{r_0} = \frac{0}{r_0} = 0$$

$$\rightarrow \theta_0^* = C_1 \exp(-\zeta_1^2 F_0)$$

$$\text{For surface } r^* = \frac{r_0}{r_0} = 1$$

$$\theta^* = C_1 \exp(-\zeta_1^2 F_0) J_0(\zeta_1 \cdot 1)$$

$$\text{w/ } F_0 = \frac{\alpha t}{r_0^2}, \quad \alpha = \frac{k}{\rho c}, \quad Bi = \frac{h r_0}{k} \quad \& \quad h = h_R + h_C$$

GIVEN:

$$\rho = 880 \text{ kg/m}^3$$

$$k = 0.52 \text{ W/m}\cdot\text{K}$$

$$c = 3350 \text{ J/kg}\cdot\text{K}$$

$$D = 2.54 \text{ cm} \rightarrow r_0 = 0.0127 \text{ m}$$

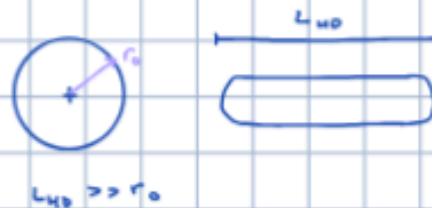
$$h_R \hat{=} 6 \text{ W/m}^2\cdot\text{K}, \quad h_C \hat{=} 12 \text{ W/m}^2\cdot\text{K}$$

$$T_i = 10^\circ\text{C} = 283.15^\circ\text{K}$$

$$T_m = 250^\circ\text{C} = 523.15^\circ\text{K}$$

$$T_{c,f} = 68^\circ\text{C} = 341.15^\circ\text{K}$$

SKETCH:



DELIVERABLES:

updated for actual calculations

$$Bi = \frac{h r_o}{k} = \frac{(12 + 6) \left[\frac{1}{1.1} - 1 \right] (0.0127)}{0.52 \left[\frac{1}{1.1} - 1 \right]}$$

$$Bi = 0.439$$

$$Fo = \frac{\alpha t}{r_o^2}, \quad \alpha = \frac{k}{\rho c} = \frac{0.52 \left[\frac{1}{1.1} - 1 \right]}{880 \left[\frac{1}{1.1} - 1 \right] + 3350 \left[\frac{1}{1.1} - 1 \right]} \quad - [m^2/s]$$

$$\alpha = 1.76 \times 10^{-7} m^2/s$$

$$Fo, \text{ centerline} = Fo, \text{ surface} = \frac{1.76 \times 10^{-7} \frac{m^2}{s}}{(0.0127 m)^2} \cdot t = 0.00109 \left[\frac{1}{s} \right] \cdot t$$

→ same calcs can be done for positions @ given times

total time to cook hot dog:

$$\theta^* = \frac{T - T_\infty}{T_i - T_\infty} = C_1 \exp(-\zeta_1^2 Fo)$$

→ updated for actual calculations

from table 5.1: using $Bi = 0.439$ (interpolating for values)

$$C_1 = 1.1014$$

$$\zeta_1 = 0.9863$$

$$\rightarrow J_0(\zeta_1) = 0.8128 \quad (\text{for final surface temp})$$

(B.4)

$$\frac{T - T_\infty}{T_i - T_\infty} = C_1 \exp(-\zeta_1^2 \cdot 0.00109 t)$$

$$\ln\left(\frac{1}{C_1} \cdot \frac{T - T_\infty}{T_i - T_\infty}\right) = -\zeta_1^2 \cdot 0.00109 t$$

$$t = \frac{-1}{0.00109 \cdot \zeta_1^2} \cdot \ln\left(\frac{1}{C_1} \cdot \frac{T_{c,f} - T_\infty}{T_i - T_\infty}\right)$$

$$= \frac{-1}{0.00109 (0.9863)^2} \cdot \ln\left(\frac{1}{1.1014} \cdot \left(\frac{341.15 - 523.15}{283.15 - 523.15}\right)\right)$$

$$t = 436.36 s = 7.27 \text{ minutes}$$

EXTRA CREDIT:

Using the non-1 term simplification,
roots are found using:

$$\sum_n \frac{J_1(\xi_n)}{J_0(\xi_n)} - B_i = 0$$

One ξ_n is found, it can be plugged into

$$C_n = \frac{2}{\xi_n} \cdot \frac{J_1(\xi_n)}{J_0^2(\xi_n) + J_1^2(\xi_n)}$$

this is done for ~15 terms.
we then use:

$$\theta^* = \frac{1 - T_n}{T_i - T_{n0}} = \sum_{n=1}^{\infty} C_n \exp(-\xi_n^2 F_0) J_0(\xi_n r^*)$$

Att. 2 Numerical Finite Differencing Development

Center line

M=Center

@ Centerpoint:

$$\Delta T \rho c \frac{(T_0^{i+1}) - T_0^i}{\Delta t} = k \left(\pi \Delta r H \right) \frac{T_1^i - T_0^i}{\Delta r}$$

$\uparrow$
 $\pi \left(\frac{\Delta r}{2} \right)^2 H$

$$\pi \left(\frac{\Delta r}{2} \right)^2 H \rho c \frac{T_0^{i+1} - T_0^i}{\Delta t} = k \left(\pi \Delta r H \right) \frac{T_1^i - T_0^i}{\Delta r}$$

$$\left(\frac{\Delta r}{2} \right)^2 \rho c \frac{1}{\Delta t} (T_0^{i+1} - T_0^i) = k (T_1^i - T_0^i)$$

$$T_0^{i+1} = \frac{k(T_1^i - T_0^i)}{\left(\frac{\Delta r}{2} \right)^2 \rho c \frac{1}{\Delta t}} - T_0^i \rightarrow T_0^{i+1} = 4 \overbrace{\frac{k \Delta t}{\rho c \Delta r^2}}^{F_0} (T_1^i - T_0^i) + T_0^i$$

$$T_0^{i+1} = 4 F_0 (T_1^i - T_0^i) + T_0^i$$

$$T_0^{i+1} = 4 F_0 T_1^i - 4 F_0 T_0^i + T_0^i$$

$$T_0^{i+1} = 4 F_0 T_1^i + T_0^i (1 - 4 F_0)$$

$$T_0^i (1 - 4 F_0)$$

$$0 < 1 - 4 F_0$$

$$4 F_0 < 1$$

$$F_0 < \frac{1}{4}, \text{ most-restrictive}$$

For Surface

$$k2\pi(M-\frac{1}{2})\Delta r H \frac{(T_{M-1}^i - T_M^i)}{\Delta r} + h[2\pi M \Delta r (H)(T_\infty^i - T_M^i)] = \underset{\substack{\uparrow \\ \pi M \Delta r H \frac{\Delta r}{2}}}{\Delta r g_c} \frac{T_M^{i+1} - T_M^i}{\Delta t}$$

$$k2\pi(M-\frac{1}{2})\Delta r H \frac{(T_{M-1}^i - T_M^i)}{\Delta r} + h[2\pi M \Delta r (H)(T_\infty^i - T_M^i)] = (\pi M \Delta r H \frac{\Delta r}{2}) g_c \frac{T_M^{i+1} - T_M^i}{\Delta t}$$

$$2k(M-\frac{1}{2})(T_{M-1}^i - T_M^i) + h(2M\Delta r)(T_\infty^i - T_M^i) = \frac{M}{2} \cdot \frac{\Delta r^2 g_c}{\Delta t} (T_M^{i+1} - T_M^i)$$

$$\frac{2}{M} \left(\frac{\Delta t}{\Delta r^2 g_c} \right) \left[2k(M-\frac{1}{2})(T_{M-1}^i - T_M^i) + h(2M\Delta r)(T_\infty^i - T_M^i) \right] = T_M^{i+1} - T_M^i$$

$$\frac{2}{M} \left(\frac{\Delta t}{\Delta r^2 g_c} \right) \left[2k \left[MT_{M-1}^i - MT_M^i - \frac{1}{2}T_{M-1}^i + \frac{1}{2}T_M^i \right] + h(2M\Delta r)(T_\infty^i - T_M^i) \right] = T_M^{i+1} - T_M^i$$

$$\frac{4F_0}{M} \left[MT_{M-1}^i - MT_M^i - \frac{1}{2}T_{M-1}^i + \frac{1}{2}T_M^i \right] + \frac{2}{M} \left(\frac{\Delta t}{\Delta r^2 g_c} \right) h(2M\Delta r)(T_\infty^i - T_M^i) =$$

$$4F_0 T_{M-1}^i - 4F_0 T_M^i - \frac{2F_0}{M} T_{M-1}^i + \frac{2F_0}{M} T_M^i + \left(\frac{4\Delta t}{\Delta r g_c} \right) h(T_\infty^i - T_M^i) + T_M^i = T_M^{i+1}$$

$$\boxed{(F_0)(T_{M-1}^i) \left(4 - \frac{2}{M} \right) + T_M^i \left(1 + \frac{2F_0}{M} - 4F_0 \right) + \left(\frac{4\Delta t}{\Delta r g_c} \right) h(T_\infty^i - T_M^i) = T_M^{i+1}}$$

$$T_M^i \left(1 + \frac{2F_0}{M} - 4F_0 \right) \quad w/M=30 \rightarrow \boxed{T_M^i (1 - 3.96F_0)}$$

$$0 < 1 - 3.96F_0$$

$$3.96F_0 < 1$$

$$\boxed{F_0 < 0.2525}$$

For m-nodes:

$$k \cancel{2r} (m - \frac{1}{2}) \cancel{\Delta r} H \left(\frac{T_{m-1}^i - T_m^i}{\cancel{\Delta r}} \right) + k \cancel{2r} (m + \frac{1}{2}) \cancel{\Delta r} H \frac{T_{m+1}^i - T_m^i}{\cancel{\Delta r}} = \cancel{2\pi m \Delta r \Delta r} H \frac{T_m^{i+1} - T_m^i}{\Delta t} \cancel{9C}$$

$$k(m - \frac{1}{2})(T_{m-1}^i - T_m^i) + k(m + \frac{1}{2})(T_{m+1}^i - T_m^i) = \frac{9C \Delta r^2}{\Delta t} (T_m^{i+1} - T_m^i)$$

$$\frac{k \Delta t}{9C \Delta r^2 m} \left[(m - \frac{1}{2})(T_{m-1}^i - T_m^i) + (m + \frac{1}{2})(T_{m+1}^i - T_m^i) \right] + T_m^i = T_m^{i+1}$$

$$F_0 \cdot \frac{1}{m} \left[\cancel{m T_{m-1}^i} - \cancel{m T_m^i} - \frac{1}{2} \cancel{T_{m-1}^i} + \frac{1}{2} \cancel{T_m^i} \right] + \left[\cancel{m T_{m+1}^i} - \cancel{m T_m^i} + \frac{1}{2} \cancel{T_{m+1}^i} - \frac{1}{2} \cancel{T_m^i} \right] + T_m^i$$

$$F_0 \left(\frac{1}{m} \right) \left[\cancel{m T_{m-1}^i} - \frac{1}{2} \cancel{T_{m-1}^i} + \cancel{m T_{m+1}^i} + \frac{1}{2} \cancel{T_{m+1}^i} - 2m T_m^i \right] + T_m^i$$

$$F_0 \left(\frac{1}{m} \right) \left[T_{m-1}^i (m - \frac{1}{2}) + T_{m+1}^i (m + \frac{1}{2}) - 2m T_m^i \right] + T_m^i$$

$$F_0 \left(\frac{1}{m} \right) \left[T_{m-1}^i (m - \frac{1}{2}) + T_{m+1}^i (m + \frac{1}{2}) \right] - 2 F_0 T_m^i \left(F_0 \left(\frac{1}{m} \right) \right) + T_m^i$$

$$F_0 \left(\frac{1}{m} \right) \left[T_{m-1}^i (m - \frac{1}{2}) + T_{m+1}^i (m + \frac{1}{2}) \right] + T_m^i (1 - 2F_0) = T_m^{i+1}$$

$$T_m^i (1 - 2F_0)$$

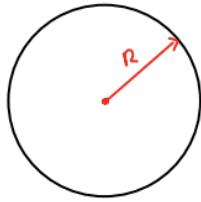
$$0 < 1 - 2F_0$$

$$2F_0 < 1$$

$$F_0 < \frac{1}{2}$$

Att. 3 Updated Heat Transfer Coefficient Development

Hot Dog



$T_\infty,$
 h_{tot}

we want $q = h A_s (T_\infty - T_s)$ (2)

↓

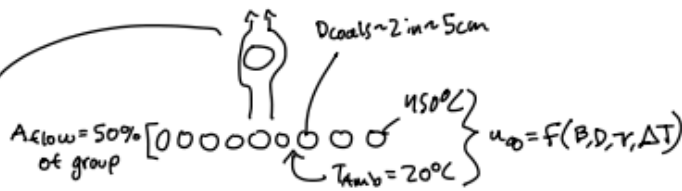
$$q = q_{tot} = q_{conv} + q_{rad}$$

$$q_{Tot} = h_{tot} A_s (T_\infty - T_s)$$

$$= h_{conv} A_s (T_\infty - T_s) + h_{rad} A_s (T_\infty - T_s)$$

we know $q_{conv} = h_{conv} A_s (T_\infty - T_s)$

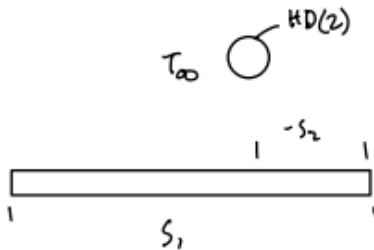
- Chapter 7: $Nu \rightarrow Pr$ (velocity comes from free convection of coals)



$$u_{amb, HD} A_{HD} = u_{amb, coals} A_{grill} = (u_{amb, coals}) (0.5) A_{grill}$$

Now we know $q_{conv} = h_{conv} A_s (T_\infty - T_s)$

$q_{rad} = ?$



Assume $T_{amb} \ll T_{coals}$
And $T_s \ll T_{coals}$
↓
 $g \ll \epsilon \epsilon_b$
And $J_{coals} = \epsilon \epsilon_b$

then $q_{1-2} = A_1 F_{12} \epsilon_1 \sigma T_1^4$

Amount absorbed $q_{ABS} = \alpha_2 \epsilon_1 A_1 F_{1-2} \sigma T_1^4$

For a diffuse, gray hot dog $\rightarrow \epsilon_2 = \alpha_2$ and using refrigerant

$$q_{ABS} = \epsilon_1 \epsilon_2 A_2 F_{2-1} \sigma T_1^4$$

Now emission from hot dog is over entire surface

$$q_{EMIT} = \epsilon_2 A_2 \sigma T_2^4$$

Multiply/Divide by $\epsilon_1 F_{21}$

$$q_{EMIT} = \epsilon_1 \epsilon_2 A_2 F_{21} \sigma \left[\frac{T_2}{(\epsilon_1 F_{21})^{1/4}} \right]^4$$

Define $T_2^* = \frac{T_2}{(\epsilon_1 F_{21})^{1/4}}$

Finally $q_{rad} = q_{abs} - q_{emit}$ Linearize
 $= \epsilon_1 \epsilon_2 A_2 F_{2-1} \sigma [T_1^4 - T_2^{*4}]$

$$q_{rad} = \epsilon_1 \epsilon_2 A_2 F_{2-1} \sigma [T_1 + T_2^*][T_1^2 + T_2^{*2}][T_1 - T_2^*]$$

h_{rad}
 $\hookrightarrow$ Unfortunately, we need $T_2^* = T_{SHD}^*$ to get h_{rad}

$$T_{S,HD}^* = f(T_{S,HD})$$

$\downarrow$

(an approximate A_s)

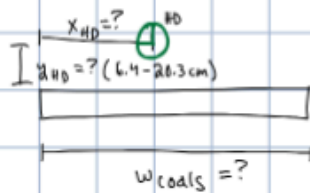
$$T_s \approx \bar{T}_s = \frac{10 + 100}{2} = 55^\circ\text{C}$$

From those experiments, we can approximate

$$T_1 - T_2^* \sim T_s - T_\infty$$

$\hookrightarrow T_{EMIT}$ of hot gases @ HD

$$q_{Tot} = (h_{conv} + h_{rad}) A_s (T_\infty - T_s)$$



$$\rightarrow X_{HD} = \frac{1}{2} W_{coals} = 10''$$

$$W_{coals} = 20''$$

$$D_{HD} = 20.3 \text{ cm} \approx 7.99''$$

$$\epsilon_1 = 0.6$$

$$\epsilon_2 = 0.45$$

$$T_{air} = 250^\circ\text{C}$$

$$T_{coals} = 450^\circ\text{C}$$

$$\sigma = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2\text{K}}$$

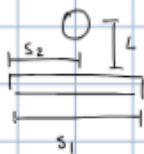
$$q_{rad} = \epsilon_1 \epsilon_2 A_2 f_{2-1} \sigma [T_1 + T_2^*] [T_1^2 + T_2^{*2}] [T_1 - T_2^*]$$

$$\rightarrow h = \epsilon_1 \epsilon_2 f_{2-1} \sigma [T_1 + T_2^*] [T_1^2 + T_2^{*2}]$$

$$\text{let } T_2^* = T_{SHD}^* \approx \frac{10 + 100}{2} = 55^\circ\text{C}$$

$$T_1 = T_{coals} = 450^\circ\text{C}$$

Approximate as cylinder & parallel rectangle



$$\left. \begin{aligned} w/S_2 &= X_{HD} \\ S_1 &= W_{coals} \\ L &= Y_{HD} \end{aligned} \right\}$$

$$f_{12} = \frac{r}{S_1 - S_2} \left[\tan^{-1} \left(\frac{S_1}{L} \right) - \tan^{-1} \left(\frac{S_2}{L} \right) \right]$$

$$= \frac{0.5''}{20'' - 10''} \left[\tan^{-1} \left(\frac{20}{7.99} \right) - \tan^{-1} \left(\frac{10}{7.99} \right) \right] = 0.892$$

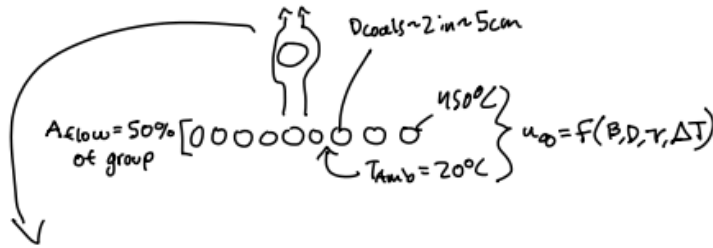
$$A_1 f_{12} = A_2 f_{21}$$

$$f_{21} = \frac{A_1}{A_2} f_{12} = \frac{W_{coals} (\text{unit depth})}{\pi D_{HD} (\text{unit depth})} f_{12}$$

$$\rightarrow h = (0.6)(0.45)(5.67 \times 10^{-8})(0.406) [(55 + 273.15) + (450 + 273.15)] [(55 + 273.15)^2 + (450 + 273.15)^2]$$

$$h_{rad} = 5.5 \frac{\text{W}}{\text{m}^2\text{K}}$$

h_{conv} : 3.11



$$u_{\infty, HD} A_{HD} = u_{\infty, coals} A_{flow} = (u_{\infty, coals}) (0.5) A_{grill}$$

u_∞ :

$$u_\infty = \sqrt{g \beta (T_s - T_\infty) D} = \sqrt{(9.81 \text{ m/s}^2) \left(\frac{1}{623 \text{ K}} \right) (1) (250 - 450 \text{ K}) (0.0254 \text{ m})} = \underline{0.2828 \text{ m/s}}$$

β :

$$\frac{450^\circ\text{C} + 250^\circ\text{C}}{2} = 350 + 273 = 623 \text{ K}$$

$$\beta = \frac{1}{623 \text{ K}}$$

Re :

$$Re_D = \frac{u_\infty D}{\nu} = \frac{(0.2828 \text{ m/s}) (0.0254 \text{ m})}{(56.15 \times 10^{-6} \text{ m}^2/\text{s})} = \underline{127.927} \rightarrow \text{laminar}$$

$$\nu: \frac{60.21 - 52.69}{650 - 600} = \frac{x - 52.69}{623 - 600} \rightarrow \nu = 56.15 \times 10^{-6} \text{ m}^2/\text{s}$$

Nu :

$$Nu_D = C Re_D^m Pr^{1/3} \text{ (for cylinder)}$$

$$Pr: \frac{0.690 - 0.685}{650 - 600} = \frac{x - 0.685}{623 - 600}$$

$$C = 0.683, m = 0.466$$

$$\hookrightarrow Pr = 0.6873$$

$$Nu_D = (0.683) (127.927)^{0.466} (0.6873)^{1/3} = \underline{5.7807}$$

$$h: \frac{Nu_D \cdot k}{D} = \frac{(5.7807) (48.188 \times 10^{-3} \text{ W/m} \cdot \text{K})}{(0.0254 \text{ m})} \quad k: \frac{49.7 - 46.9}{650 - 600} = \frac{x - 46.9}{623 - 600}$$

$$\boxed{h_{conv} = 10.9669 \text{ W/m}^2 \cdot \text{K}}$$

$$k = 48.188 \text{ W/m} \cdot \text{K}$$

Att. 4 MATLAB CODE

Analytical Solution:

```
%% Plots of Analytical Hot Dog Temp. Solutions

% Clean up command window
clear;
close all;
clc;

% Set up givens and pre-calculated constants
rho = 880;           % [kg/m^3]
k = 0.52;            % [W/m-K]
c = 3350;            % [J/kg-K]
D = 2.54E-2;         % [m]
r0 = D/2;           % [m]
% hr = 2.29;         % [W/m^2-K]
% hc = 10.96;        % [W/m^2-K]
% h = hc + hr;       % [W/m^2-K]
Ti = 283.15 - 273.15; % [K]
Tinf = 523.15 - 273.15; % [K]
Tcf = 341.15 - 273.15; % [K]

%for hrad = 2.29
% C1 = 1.0712;
% zeta1 = 0.7465;

%for hrad = 5.5, Bi 0.402
C1 = 1.0936;
zeta1 = 0.8534;

Fourier_Coeff = 0.00109;
bessel_zeta1 = besselj(0,zeta1);

t_total = 463;      % [seconds]
t1 = 30;            % [seconds]
t2 = 120;           % [seconds]

% Set up time and position vectors
t = 0:0.5:t_total;
r = 0:0.0001:r0;
rstar = r/r0;

% Solve for temps vs. time
T_Center = Tinf + (Ti - Tinf).*C1.*exp(-zeta1^2.*0.00109.*t);
T_Surface = Tinf + (Ti - Tinf).*C1.*exp(-zeta1^2.*0.00109.*t).*bessel_zeta1;

% Solve for temps vs. position
% Solve for constants that don't change with position for clarity of code
T30_coeff = (Ti - Tinf)*C1*exp(-zeta1^2*0.00109*t1);
T120_coeff = (Ti - Tinf)*C1*exp(-zeta1^2*0.00109*t2);
Tfinals_coeff = (Ti - Tinf)*C1*exp(-zeta1^2*0.00109*t_total);

T30s = Tinf + T30_coeff.*besselj(0,zeta1.*(rstar));
T120s = Tinf + T120_coeff.*besselj(0,zeta1.*(rstar));
Tfinals = Tinf + Tfinals_coeff.*besselj(0,zeta1.*(rstar));

% Plot Results

% Temp Vs. Time
figure(1);
```

```

hold on
grid on
plot(t,T_Center,'r')
plot(t,T_Surface,'b')
hold on
title('Temperature of Hot Dog Vs. Time')
xlabel('Time (Seconds)')
ylabel('Temperature (Celcius)')

% Temp vs. Position
figure(2);
hold on
grid on
plot(r,T30s,'r')
plot(r,T120s,'b')
plot(r,Tfinals,'g')
hold on
title('Temperature of Hot Dog Vs. Position')
xlabel('Position (Meters)')
ylabel('Temperature (Celcius)')

%% Extra Credit 15 Terms
%Temp vs. Time

Bi = 0.402;
myfun = @(z) z*(besselj(1,z)/besselj(0,z)) - Bi;
zeta = zeros(1,15);
coeffs = zeros(1,15);
guess = 0.8;

temp = 0;

for k = 1:15
    zeta(k) = fzero(myfun,guess);
    guess = guess + pi;

    num = besselj(1,zeta(k));
    denom = (besselj(0,zeta(k)))^2+(besselj(1,zeta(k)))^2;
    coeffs(k) = (2/zeta(k))*(num/denom);
end

% Centerline vs. Time
temp_center = zeros(873,1);
counter = 0;

for t_iterate = 0:0.5:t_total

    Te_Center = (Ti - Tinf).*coeffs.*exp(-zeta.^2.*0.00109.*t_iterate);

    counter = counter + 1;
    temp_center(counter) = sum(Te_Center) + 250;
end

% Surface vs. Time
temp_surface = zeros(873,1);
counter = 0;

```

```

for t_iterate = 0:0.5:t_total
    Te_Surface = (Ti - Tinf).*coeffs.*exp(-zeta.^2*0.00109.*t_iterate).*besselj(0,zeta);
    counter = counter + 1;
    temp_surface(counter) = sum(Te_Surface) + 250;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Temp vs. Position

T30_coeff = (Ti - Tinf).*coeffs.*exp(-zeta.^2*0.00109*t1);
T120_coeff = (Ti - Tinf).*coeffs.*exp(-zeta.^2*0.00109*t2);
Tfinals_coeff = (Ti - Tinf).*coeffs.*exp(-zeta.^2*0.00109*t_total);

%t30
position_T30 = zeros(128,1);
counter = 0;
for r_iterate = 0:0.0001:r0
    T30s_exact = T30_coeff.*besselj(0,zeta1.*(r_iterate/r0));

    counter = counter + 1;
    position_T30(counter) = sum(T30s_exact) + Tinf;
end

%t120
position_T120 = zeros(128,1);
counter = 0;
for r_iterate = 0:0.0001:r0
    T120s_exact = T120_coeff.*besselj(0,zeta1.*(r_iterate/r0));

    counter = counter + 1;
    position_T120(counter) = sum(T120s_exact) + Tinf;
end

%tfinal
position_Tfinals = zeros(128,1);
counter = 0;
for r_iterate = 0:0.0001:r0
    Tfinals_exact = Tfinals_coeff.*besselj(0,zeta1.*(r_iterate/r0));

    counter = counter + 1;
    position_Tfinals(counter) = sum(Tfinals_exact) + Tinf;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%Graphs

% Vs. Time

figure(1)
plot(t,temp_center,'m')
plot(t,temp_surface,'c')
legend('Centerline Temperature','Surface Temperature','Centerline Temperature Exact', ...

```

```

        'Surface Temperature Exact','Location','northwest')

% Vs. Position

figure(2)
plot(r,position_T30,'m');
plot(r,position_T120,'c');
plot(r,position_Tfinals,'y');
legend('30 Seconds Temp','120 Seconds Temp', 'Fully Cooked Temp', ...
        '30 Seconds Temp Exact','120 Seconds Temp Exact', 'Fully Cooked Temp Exact','Location','northwest')

```

Numerical Finite difference solution:

```

%% Finite Hot Dog Solution

%% Set-up %%
clc;
clear;
close all;

% Set up Constants
rho = 880; % kg/m^3
k = 0.52; % W/m-K
c = 3350; % J/kg-K
D = 0.0251618254; % m
r = D/2; % m

T_inf = 250; % C
h_conv = 10.96; % W/m^2-K
h_rad = 5.5; % W/m^2-K
h_tot = h_rad + h_conv;

% Set up Steps (Temp)
Nodes = 30;
delta_r = r/Nodes; %m
delta_t = (1/4*rho*c*delta_r^2)/k;
Fo = 1/4;

% Set up Temps
t_total = 450; % [seconds]
t1 = 30; % [seconds]
t2 = 120; % [seconds]

T = zeros(Nodes,length(0:delta_t:t_total));
T(:,1) = 10; %initial temp in hot dog is 10°C

%% Temps Vs. Time
%Solve Temps

for k = 1:1:length(1:delta_t:t_total)

    if T(1,k) >= 68
        break
    end

    %Centerline
    %T(1,k+1) = (1/4)*((k*delta_t)/(rho*c*delta_r^2))*(T(2,k)-T(1,k))+T(1,k);
    T(1,k+1) = 4*Fo*T(2,k)+T(1,k)*(1-4*Fo);
    for j = 2:29
        %M1-30 Nodes
        %T(j,k+1) = ((k*delta_t)/(j*rho*c*delta_r^2))*((j-0.5)*(T(j-1,k)-T(j,k)))+(j+0.5)*(T(j+1,k)-T(j,k))) + T(j,k);
        %T(j,k+1) = (1/j)*Fo*((j-0.5)*(T(j-1,k)-T(j,k)))+(j+0.5)*(T(j+1,k)-T(j,k))) + T(j,k);
        T(j,k+1) = Fo*(1/j)*(T(j-1,k)*(j-0.5)+T(j+1,k)*(j+0.5)) + T(j,k)*(1-2*Fo);
    end

    %Surface
    %T(30,k+1) = (((1/30)*2*delta_t)/(rho*c*delta_r^2))*((2*k*(30-0.5))*(T(29,k)-T(30,k)))+7*(2*30*delta_r)*(T_inf-T(30,k))+T(30,k);
    %T(30,k+1) = (1/Nodes)*2*Fo*((30-0.5)*(T(29,k)-T(30,k))+h_tot*((2*30*delta_r)*(T_inf-T(30,k)))+T(30,k);

```

```

        T(30,k+1) = Fo*(T(Nodes-1,k))*(4-(2/Nodes)) + T(Nodes,k)*(1+((2*Fo)/Nodes)-4*Fo) + k
        h_tot*(4*delta_t/(delta_r*rho*c))*(T_inf-T(Nodes,k));
    end

    time = delta_t*(1:1:length(T));
    cook_time = k*delta_t;

%% Plots
figure()
plot(time,T(1,:), 'x-');

hold on
grid on

plot(time,T(30,:), 'x-')
xlim([0 cook_time])
title('Hot Dog Temperature Vs. Time')
xlabel('Time (Seconds)')
ylabel('Temperature (Celcius)')
legend('Centerline', 'Surface', "Location", 'northwest')

txt = ['Total Time Elapsed: ' num2str(cook_time) ' seconds'];
text(190,23,txt)

hold off
%% Temps Vs. Position
% Just Need to Plot them from respective columns

figure();
hold on
grid on

%THESE TIMES ARE WRONG, NEED TO CALC DELTA T FIRST

x = delta_r*(0:1:29);

sec_30 = round(30/delta_t);
sec_120 = round(120/delta_t);

% 30 Seconds
plot(x,T(:,sec_30), 'x-')
% 120 Seconds
plot(x,T(:,sec_120), 'x-')
% Fully Cooked
plot(x,T(:,k), 'x-')

ylim([0 120])
xlim([0 0.01216])

title('Hot Dog Temperature Vs. Position')
xlabel('Radial Position (m)')
ylabel('Temperature (Celcius)')
legend('30 Seconds', '120 Seconds', 'Fully Cooked', "Location", 'northwest')

```